Zida Li, Ph.D.

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Biomedical Engineering, Shenzhen University
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Education

University of Michigan, Ann Arbor (UM) — Ann Arbor, MI, US

Ph.D., Mechanical Engineering — Aug. 2013 – Apr. 2018

Dissertation: *Micro-Engineered Devices for Point-of-Care Blood Clot Retraction Testing*

Advisor: Prof. Jianping Fu

University of Science and Technology of China (USTC) — Hefei, Anhui, China

B.Eng., Mechanical Engineering — Aug. 2008 – June 2012

Advisor: Prof. Liqun He

Tsinghua University — Beijing, China

Exchange Program – C9 University League — Sept. 2010 – Feb. 2011

Positions and Employment

Shenzhen University (SZU) — Shenzhen, China

Assistant Professor, Biomedical Engineering — June 2018 – present

University of Michigan, Ann Arbor — Ann Arbor, MI, US

Graduate Student Research Assistant, Mechanical Engineering — Sept. 2013 – Apr. 2018

Graduate Student Teaching Assistant, Mechanical Engineering — Sept. 2014 – Apr. 2018

University of Hong Kong — Hong Kong

Research Assistant, Mechanical Engineering — Aug. 2012 – June 2013

Advisor: Prof. Anderson Ho Cheung Shum

Honors and Awards

- Outstanding Undergrad Mentor Award, SZU (2022)
- Outstanding Undergrad Instructor Award, SZU (2022)
- Tier Three Award in Equipment Design for Laboratory Classes, 6th National Competition of Teaching Innovation, Chinese Association of Higher Education, Ministry of Education, China (2021)
- Advisor Award for Distinguished Undergrad Thesis (advisee: Meichi Jin), SZU (2021)
- University Teaching Award, SZU (2021)
- Excellence in Faculty Performance Evaluation, SZU (2020)
- Baxter Young Investigator Award First-Tier, Baxter Healthcare Inc. (2016)
- Provincial Honored Graduate, Department of Education, Anhui Province, China (2012)
- National Scholarship, Ministry of Education, China (2011)

- National Encouragement Scholarship, Ministry of Education, China (2010)
- Qian Jun Scholarship, USTC (2009)

Research Grants

- Early Career Grant (青年基金), National Natural Science Foundation of China (PI, ￥300K, 2024-2026)
- Industrial collaboration grant for research in single-cell RNA sequencing (PI, ￥300K, 2022-2023)
- Junior Faculty Development Award, Department of Biomedical Engineering, SZU (PI; ￥100K, 2019)
- General Grant (面上基金), Science and Technology Agency, Guangdong (PI, ￥100K, 2019-2021)
- Grant for Research in Medical Science, Committee of Hygiene and Health, Guangdong (PI, ￥10K, 2019-2021)
- Faculty Startup Grant, Shenzhen (PI, ￥2.7M, 2020-2022)
- Faculty Startup Grant, SZU (PI, ￥200K, 2019-2022)

Selected Publications

[#]first authors; *corresponding author(s); underscore: student advisees.

[1] Linzhe Chen,[#] Jingyi Ding,[#] Hao Yuan, Chi Chen*, and **Zida Li*** (2022). deep-dLAMP: deep learning-enabled polydisperse emulsion-based digital loop-mediated isothermal amplification. ***Advanced Science***, 9(9), 2105450

[2] Kai Wu,[#] Qi Fang,[#] Zhantao Zhao, and **Zida Li*** (2023). CoID-LAMP: Color-encoded, intelligent digital LAMP for multiplexed nucleic acid quantification. ***Analytical Chemistry***, 95(11), 5069–5078

[3] Run Xie,[#] Yang Liu,[#] Xuyang Shi, Shiyu Wang, Zhantao Zhao, Longqi Liu, Ya Liu,* and **Zida Li*** (2023). Combinatorial perturbation sequencing on single cells using microwell-based droplet random pairing. ***Biosensors & Bioelectronics***, 220, 114913

[4] Linzhe Chen, Donghao Li, Xinyu Liu, Yihan Xie, Jieying Shan, Haofan Huang, Xiaxia Yu, Yudan Chen, Weidong Zheng, and **Zida Li*** (2022). Point-of-care blood coagulation assay based on dynamic monitoring of blood viscosity using droplet microfluidics. ***ACS Sensors***, 7(8), 2170–2177

[5] Donghao Li,[#] Xinyu Liu,[#] Yujuan Chai,[#] Jieying Shan, Yihan Xie, Yong Liang, Susu Huang, Weidong Zheng, and **Zida Li*** (2022). Point-of-care blood coagulation assay enabled by printed circuit board-based digital microfluidics. ***Lab on a Chip***, 22(4), 1473-0197

[6] Lanzhu Huang,[#] Xinyu Liu,[#] Yuanbin Ou, Haofan Huang, Xia Zhang, Yize Wang, Yong Liang, Xiaxia Yu, Weidong Zheng, Huisheng Zhang, and **Zida Li*** (2020). Micro-engineered flexural post rings for effective blood sample fencing and high throughput measurement of clot retraction force. ***ACS Sensors***, 5(12), 3949-3955

[7] Meichi Jin, Kai Wu, Mengzhen Wang, Yang Zhang, Chengbin Yang, and **Zida Li*** (2023). High resolution, multiplex antibody patterning using micropillar-focused droplet printing and microcontact printing. ***Advanced Biology***, in press, DOI: 10.1002/adbi.202300111

Refereed Journal Publications (full list)

[#]first authors; *corresponding author(s); underscore: student advisees.

[1] Meichi Jin, Kai Wu, Mengzhen Wang, Yang Zhang, Chengbin Yang, and **Zida Li*** (2023). High resolution, multiplex antibody patterning using micropillar-focused droplet printing and microcontact printing. ***Advanced Biology***, 7(8), 2300111, 2023

[2] Kai Wu,[#] Qi Fang,[#] Zhantao Zhao, and **Zida Li*** (2023). CoID-LAMP: Color-encoded, intelligent digital LAMP for multiplexed nucleic acid quantification. ***Analytical Chemistry***, 95(11), 5069–5078

[3] Run Xie,[#] Yang Liu,[#] Xuyang Shi, Shiyu Wang, Zhantao Zhao, Longqi Liu, Ya Liu,* and **Zida Li*** (2023). Combinatorial perturbation sequencing on single cells using microwell-based droplet random pairing. ***Biosensors & Bioelectronics***, 220, 114913

[4] Yang Liu,[#] Shiyu Wang,[#] Menghua Lyu,[#] Run Xie, Weijin Guo, Ying He, Xuyang Shi, Yang Wang, Jingyu Qi, Qianqian Zhu, Hui Zhang, Tao Luo, Huaying Chen, Yonggang Zhu, Xuan Dong, **Zida Li**, Ying Gu, Feng Mu, Longqi Liu,* Xun Xu,* and Ya Liu* (2022). Droplet microfluidics forward for tracing target cells at single-cell transcriptome resolution. ***Bioengineering***, 9(11), 674

[5] Yang Zhang, Taozhao Yu, Jingyi Ding, and **Zida Li*** (2023). Bone-on-a-chip platforms and integrated biosensors: towards advanced *in vitro* bone models with real-time biosensing. ***Biosensors & Bioelectronics***, 219, 114798

[6] Menghua Lyu,[#] Xuyang Shi,[#] Xiaopan Liu,[#] Xijun Zhu, Yang Liu, Lijuan Liao, Shiyu Wang, Na Sun, Hongyan Zhao, Linzhe Chen, Linyuan Fan, Qumiao Xu, Qianqian Zhu, Kai Gao, Huaying Chen, Yonggang Zhu, **Zida Li**, Weijin Guo, Yue Zheng, Ying Gu, Longqi Liu,* Meiniang Wang,* and Ya Liu* (2022). Generation and screening of antigen-specific nanobodies from mammalian cells expressing BCR repertoire library using droplet-based microfluidics. ***Analytical Chemistry***, 94(22), 7970–7980, 2022

[7] Linzhe Chen, Donghao Li, Xinyu Liu, Yihan Xie, Jieying Shan, Haofan Huang, Xiaxia Yu, Yudan Chen, Weidong Zheng, and **Zida Li*** (2022). Point-of-care blood coagulation assay based on dynamic monitoring of blood viscosity using droplet microfluidics. ***ACS Sensors***, 7(8), 2170–2177

- Selected as Front Cover story by *ACS Sensors*

[8] Linzhe Chen,[#] Jingyi Ding,[#] Hao Yuan, Chi Chen*, and **Zida Li*** (2022). deep-dLAMP: deep learning-enabled polydisperse emulsion-based digital loop-mediated isothermal amplification. ***Advanced Science***, 9(9), 2105450

[9] Donghao Li,[#] Xinyu Liu,[#] Yujuan Chai,[#] Jieying Shan, Yihan Xie, Yong Liang, Susu Huang, Weidong Zheng, and **Zida Li*** (2022). Point-of-care blood coagulation assay enabled by printed circuit board-based digital microfluidics. ***Lab on a Chip***, 22(4), 1473-0197

[10] **Zida Li**,[#,*] Feng Lin,[#] Shue Wang, Xufeng Xue, and Yue Shao* (2022). Single-cell sequencing to unveil the mystery of embryonic development. ***Advanced Biology***, 6(2), 2701-0198

[11] Shiyu Wang,[#] Yang Liu,[#] Yijian Li, Menghua Lv, Kai Gao, Ying He, Wenbo Wei, Yonggang Zhu, Xuan Dong, Xun Xu, **Zida Li**,* Longqi Liu,* and Ya Liu* (2022). High-throughput functional screening of antigen-specific T-cells based on droplet microfluidics on single-cell level. ***Analytical Chemistry***, 94(2), 918–926

- Selected as Front Cover story by *Analytical Chemistry*

[12] Linzhe Chen, Guoliang Zhang, Longqi Liu,* and **Zida Li*** (2021). Emerging biosensing technologies for improved diagnostics of COVID-19 and future pandemics. ***Talanta***, 225, 121986

- ESI Highly Cited Paper in the academic field of Chemistry as of July/August 2021

[13] Lanzhu Huang,[#] Xinyu Liu,[#] Yuanbin Ou, Haofan Huang, Xia Zhang, Yize Wang, Yong Liang, Xiaxia Yu, Weidong Zheng, Huisheng Zhang, and **Zida Li*** (2020). Micro-engineered flexural post rings for effective blood sample fencing and high throughput measurement of clot retraction force. ***ACS Sensors***, 5(12), 3949-3955

- Selected as Front Cover story by *ACS Sensors*
- Highlighted in Introducing Our Authors by *ACS Sensors* (2020, 5(12), 3653–3654)

[14] Zhourui Xu, **Zida Li**, Yihang Jiang, Gaixia Xu, Mingwei Zhu, Wing-Cheung Law, Ken-Tye Yong, Yanshuai Wang, Chengbin Yang, Biqin Dong, and Feng Xing* (2020). Recent advances in solar-driven evaporation system. ***Journal of Materials Chemistry A***, 8, 25571-25600

[15] Xue Chen, Nicolo Simone Villa, Yanfeng Zhuang, Linzhe Chen, Tianfu Wang, **Zida Li**,* and Tiantian Kong* (2020). Stretchable supercapacitors as emergent energy storage units for health monitoring bioelectronics. ***Advanced Energy Materials***, 10(4), 1902769

[16] Yi Zheng, Xufeng Xue, Yue Shao, Sicong Wang, Sajedeh Nasr Esfahani, **Zida Li**, Jonathon M. Muncie, Johnathon N. Lakins, Valerie M. Weaver, Deborah L. Gumucio, and Jianping Fu* (2019). Controlled modeling of human epiblast and amnion development using stem cells. ***Nature***, 573(7774), 421-425

[17] Yuanyuan Zheng,[#] Xufeng Xue,[#] Agnes M. Resto Irizarry, **Zida Li**, Yue Shao, Yi Zheng, Gang Zhao,* and Jianping Fu* (2019). A patterned model for neural tube development studies by human embryonic stem cells in a biomimetic niche. ***Science Advances***, 5(12), eaax5993

[18] Sajedeh Nasr Esfahani, Yue Shao, Agnes M Resto Irizarry, **Zida Li**, Xufeng Xue, Deborah L Gumucio, and Jianping Fu* (2019). Microengineered human amniotic ectoderm tissue array for high-content developmental phenotyping. ***Biomaterials***, 216, 119244

[19] Luoquan Li[#], Ping Wu[#], Zhaofeng Luo, Lei Wang, Weiping Ding, Tao Wu, Jinyu Chen, Jinlong He, Ying Chen, Guibo Li, **Zida Li**,* and Liqun He* (2019). Dean flow assisted single cell and bead encapsulation for high performance single cell expression profiling. ***ACS Sensors***, 4(5), 1299-1305

[20] **Zida Li**,* Luoquan Li, Meixiang Liao, Liqun He, and Ping Wu* (2019). Multiple splitting of droplets using multi-furcating microfluidic channels. ***Biomicrofluidics***, 13(2), 024112

[21] Feng Lin, Yue Shao, Xufeng Xue, Yi Zheng, **Zida Li**, Chunyang Xiong, Jianping Fu* (2019). Biophysical phenotypes and determinants of anterior vs. posterior primitive streak cells derived from human pluripotent stem cells. ***Acta Biomaterialia***, 86, 125-134

[22] **Zida Li**, Yize Wang, Xufeng Xue, Brendan McCracken, Kevin Ward, and Jianping Fu* (2018). Carbon nanotube strain sensor based hemoretractometer for blood coagulation testing. ***ACS Sensors***, 3(3), 670-676

[23] **Zida Li**, Xufeng Xue, Feng Lin, Yize Wang, Kevin Ward, and Jianping Fu* (2017). Capillary-assisted coating of carbon nanotube thin film as a strain gauge. ***Applied Physics Letters***, 111(17), 173105

[24] Koh Meng Aw Yong, **Zida Li**, Sofia D. Merajver, and Jianping Fu* (2017). Analysis of tumor invasion front using long-term fluidic tumoroid culture. ***Scientific Reports***, 7(1), 10784

[25] Xufeng Xue, Xiaowei Hong, **Zida Li**, Cheri X. Deng, and Jianping Fu* (2017). Acoustic tweezing cytometry enhances osteogenesis of human mesenchymal stem cells through cytoskeletal contractility and YAP activation. ***Biomaterials***, 134, 22-30

[26] Jianming Sang, Xiang Li, Yue Shao, **Zida Li**, and Jianping Fu* (2016) Controlled tubular unit formation from collagen film for modular tissue engineering. ***ACS Biomaterials Science & Engineering***, 3(11), 2860-2868

[27] **Zida Li**, Xiang Li, Brendan McCracken, Yue Shao, Kevin Ward, and Jianping Fu* (2016). A miniaturized hemoretractometer for blood clot retraction testing. ***Small***, 12(29), 3926-3934.

- Selected as Frontispiece story by *Small*

[28] Ping Wu, Zhaofeng Luo, Zhifeng Liu, **Zida Li**, Chi Chen, Lili Feng, and Liqun He* (2015). Drag-induced breakup mechanism for droplet generation in dripping within flow focusing microfluidics. ***Chinese Journal of Chemical Engineering***, *23*(1), 7-14

[29] **Zida Li**, Sze Yi Mak, Alban Sauret, and Ho Cheung Shum* (2014). Syringe-pump-induced fluctuation in all-aqueous microfluidic system implications for flow rate accuracy. ***Lab on a Chip***, 14(4), 744-749

[30] Sze Yi Mak, **Zida Li**, Arnaud Frere, Tat Chuen Chan, and Ho Cheung Shum* (2014). Musical Interfaces: Visualization and Reconstruction of Music with a Microfluidic Two-Phase Flow. ***Scientific Reports***, 4, 6675

[31] Xiang Li, Weiqiang Chen, **Zida Li**, Ling Li, Hongchen Gu, and Jianping Fu* (2014). Emerging microengineered tools for functional analysis and phenotyping of blood cells. ***Trends in Biotechnology***, 32(11), 586-594

Refereed Book Chapters

[1] **Zida Li*** and Anderson Ho Cheung Shum* (2019). Nanotechnology and microfluidics for biosensing and biophysical property assessment: implications for next generation *in vitro* diagnostics. ***Nanotechnology and Microfluidics***, 83-107, John Wiley & Sons

Patents

[1] **Zida Li**, Qi Fang, and Kai Wu (2023). Method, device, and medium for multiple digital detection of nucleic acid with deep learning. *China Patent Application* ZL202211516857.X

[2] **Zida Li**, Xiaxia Yu, Xinyu Liu, Jieying Shan, and Yihan Xie (2023). Simulation system and method of in vitro diagnostics. *China Patent Application* ZL202110750662.0

[3] **Zida Li**, Lanzhu Huang, and Weidong Zheng (2022). A fabrication method and application of soft post rings for clot retraction testing. *China Patent Application* ZL202010260648.8

[4] Jianping Fu, Kevin Ward, **Zida Li**, and Xiang Li (2017). A microscale device for blood coagulation assay. *U.S. Patent Application* 62/304,385

[5] Ho Cheung Shum, Alban Sauret, **Zida Li**, and Yang Song (2013). System and method for generation of emulsions with low interfacial tension and measuring frequency of vibrations in the system. *U.S. Patent Application* 13/839,072

Conference Presentations

[1] Single-cell chemical transcriptome profiling for drug screening. **Panel speech.** *15th IEEE International Conference on Nano/Molecular Medicine & Engineering*. Online, Nov. 2021

[2] Micro-engineered devices for point-of-care blood clot retraction testing. **Panel Speech**. *3rd International Conference of Microfluidics, Nanofluidics, and Lab-on-a-Chip*, Shenzhen, China, July 2021

[3] Micro-engineered devices for point-of-care blood clot retraction testing. **Panel Speech**. *8th Conference on Micro-Total Analysis*, Shenzhen, China, Apr. 2021

[4] Micro-engineered devices for point-of-care blood clot retraction testing. **Panel Speech**. *4th Conference of Microfluidics Technology and Innovation*, Shenzhen, China, Dec. 2020

[5] Dean flow assisted single cell and bead encapsulation for high performance single cell expression profiling. **Panel Speech**. *7th Forum on Lab-on-a-Chip Applications*, Dalian, China, Nov. 2019

[6] Dean flow assisted single cell and bead encapsulation for high performance single cell expression profiling. **Panel Speech**. *9th International Multidisciplinary Conference on Optofluidics*, Hong Kong, China, June 2019

[7] Capillary-facilitated coating of carbon nanotube thin film as a strain gauge for blood retraction testing. **Poster Presentation**. *Conference of Micro-Total Analysis System*, Savannah, GA, USA, Oct. 2017

[8] Capillary-assisted coating of carbon nanotube thin film for blood retraction testing. **Panel Speech**. *Biomedical Engineering Society Annual Meeting 2017*, Phoenix, AZ, USA, Oct 2017

[9] A miniaturized hemoretractometer for blood clot retraction testing. **Panel Speech**. *8th International Symposium on Microchemistry and Microsystems*, Hong Kong, May 2016

Invited Seminars

[1] Droplet microfluidics-based nucleic acid quantification and single cell analysis. Department of Mechanical Engineering, **University of Science and Technology of China**, Hefei, China, Mar. 2023

[2] Droplet microfluidics-based nucleic acid quantification and single cell analysis. Department of Chemical Engineering, **Zhejiang University**, Hangzhou, China, Mar. 2023

[3] Point-of-care blood coagulation assays using droplet microfluidics and digital microfluidics. **magAssist Inc.**, Mar. 2023

[4] Droplet microfluidics-based nucleic acid quantification and single cell analysis. Department of Mechanical Engineering, **Northern Arizona University**, Online, Feb. 2022

[5] Single-cell analysis using microfluidics. College of Engineering, **Peking University**, Online, Nov. 2021

[6] Microfluidics-enabled point-of-care testing and single cell analysis. Department of Biomedical Engineering, **Shenzhen University**, Shenzhen, China, Dec. 2020

[7] Droplet microfluidics and single cell analysis. Department of Thermal Science and Energy Engineering, **University of Science and Technology of China**, Hefei, China, Nov. 2019

[8] Micro/Nano-engineered tools for mechanobiology. Department of Mechanical and Electrical Engineering, **Guilin University of Electronic Technology**, Guilin, China, Dec. 2018

[9] Micro-engineered blood coagulation tests. Department of Thermal Science and Energy Engineering, **University of Science and Technology of China**, Hefei, China, Mar. 2018

Student advising

- Master's students

[1] Donghao Li 李东豪 (2023). "Point-of-care blood coagulation assays using digital microfluidics."

[2] Kai Wu 武凯 (2023). "Multiplex digital LAMP using droplet color-coding and intelligent image analysis."

[3] Run Xie 谢润 (2023). "High throughput analysis of chemical transcriptomes using droplet pairing and single-cell RNA sequencing."

[4] Linzhe Chen 陈琳喆 (2022). "Point-of-care testing based on droplet microfluidics."

[5] Tao Wang 汪涛 (2022). "Single Cell Isolation Using Droplet-based Microfluidics."

[6] Lanzhu Huang 黄兰蛛 (2021). "Flexible Micropost Rings for High Throughput Testing of Clot Retraction Force."

- Bachelor's students

[1] Yihan Xie 谢以瀚 (2023). "Single-cell RNA sequencing from sorted cells with low number."

[2] Yunzhu Wan 万芸竹 (2022). "Digital nucleic acid amplification tests using non-uniform compartments."

[3] Jieying Shan 单洁滢 (2022). "Digital microfluidics and its biomedical applications."

[4] Jinying Cai 蔡金颖 (2021). "Digital microfluidics for reagent processing in *in vitro* diagnostics."

[5] Meichi Jin 金美池 (2021). "Antibody patterning using micropillar-focused droplet printing."

Teaching

[1] Calculus for Medical Students. 2018-19, 19-20, 20-21, 21-22, 22-23. Average student rating: Top 10%.

[2] Biomedical Sensors and Applications. 2018-19. Student rating: Top 10%.

[3] Single Cell Sequencing (short course). 2019-20, 20-21.

[4] Scientific Reading and Writing (Master's level). 2018-19, 19-20, 20-21, 21-22.